INCH-POUND

MIL-DTL-83513/15D
14 July 2004
SUPERSEDING
MIL-DTL-83513/15C
20 September 2002

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, RECEPTACLE, MICROMINIATURE, POLARIZED SHELL, RIGHT ANGLE, SOCKETS CONTACTS, 4 ROW, SOLDER TYPE, NARROW PROFILE, 100 CONTACTS, PRINTED CIRCUIT BOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83513.

Inches	mm
.003	0.08
.005	0.13
.010	0.25
.0150	0.38
.025	0.64
.050	1.27
.109	2.77
.125	3.18
.130	3.30
.144	3.66
.175	4.45
.185	4.70
.200	5.08
.2712	6.89
.400	10.16
.420	10.67
.590	14.99
1.010	25.65
1.3842	35.16
1.800	45.72
1.825	46.36
2.175	55.25

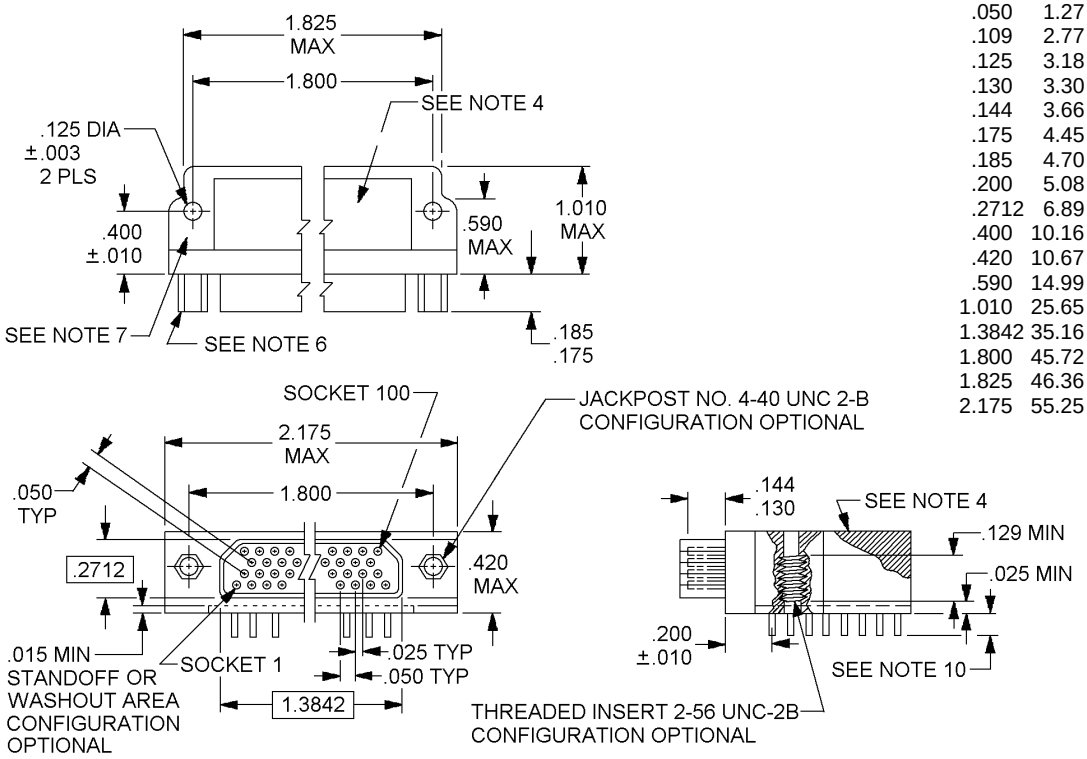
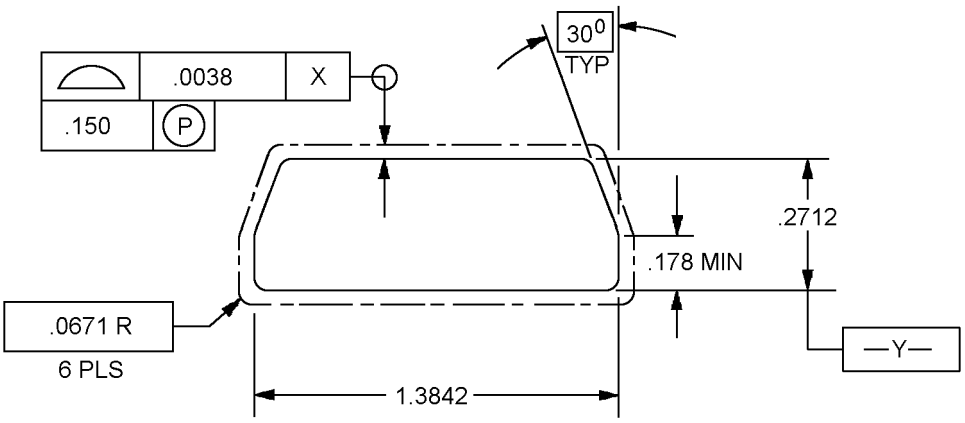


FIGURE 1. Connector, receptacle, .050 spacing.

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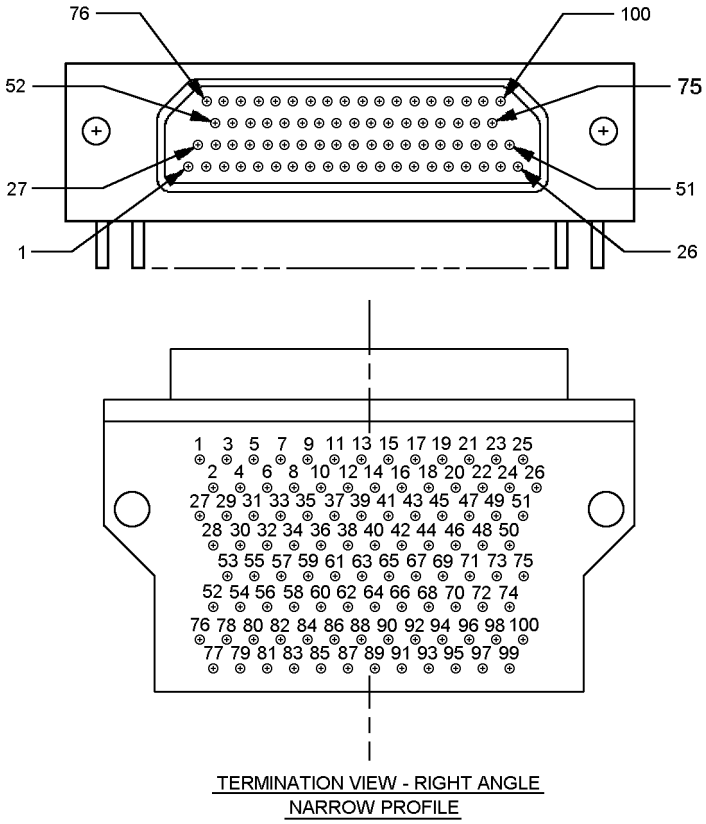
DETAIL OF INTERFACE

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ± .005 (0.13 mm).
4. Termination organization area to be optionally molded or filled with a potting fill material capable of passing the electrical and environmental requirements of MIL-DTL-83513. Plastic molding shall conform to type GDI-30F or type SDG-F in accordance with ASTM D5948 or GCT-30F in accordance with ASTM D5927 or MIL-M-24519 or GST-40F in accordance with ASTM D4067 or MIL-M-24519 or GLCP-30F or GLCP-50 in accordance with ASTM D5138 or MIL-M-24519.
5. Metal shell shall be of material in accordance with MIL-DTL-83513 for class M parts.
6. Jackpost (permanently attached), when specified: Corrosion resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with SAE-AMS-QQ-P-35.
7. Separately molded plastic body (if used) shall conform to type GDI-30F or type SDG-F in accordance with ASTM D5948 or GCT-30F in accordance with ASTM D5927 or MIL-M-24519 or GST-40F in accordance with ASTM D4067 or MIL-M-24519 or GLCP-30F or GLCP-50 in accordance with ASTM D5138 or MIL-M-24519
8. Wire termination sockets shall conform to A-A-59551, no. 24 AWG copper.
9. Interfacial seals are included with each type receptacle connector.
10. Termination lengths available: .109 (2.77 mm), .140 (3.56 mm), or .172 (4.37 mm). The tolerance shall be ± .015 (0.38 mm) for all termination lengths.
11. Threaded insert, when specified: Corrosion resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with SAE-AMS-QQ-P-35.

FIGURE 1. Connector, receptacle, .050 spacing - Continued.

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TERMINATION VIEW - RIGHT ANGLE
NARROW PROFILE

NOTE: Engaging face of socket insert shown, cavity identification numbers are for reference only and do not appear on the part.

FIGURE 2. Insert arrangement.

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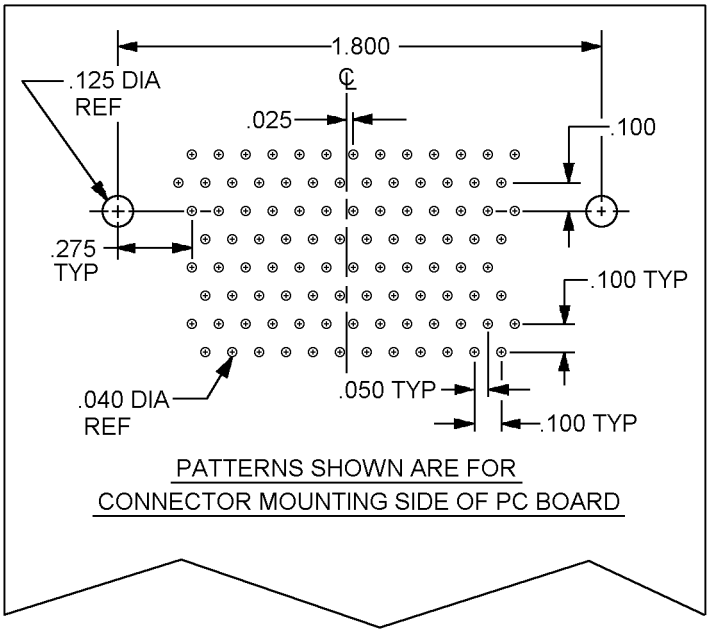


FIGURE 3. Layout arrangement.

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REQUIREMENTS

Dimensions and configurations: See figures 1, 2, and 3.

Current rating, maximum: 3 amperes per contact.

Materials:

Termination organization area: Potting fill material capable of passing the electrical and environmental requirements of MIL-DTL-83513.

Shell: The requirements for shell materials shall be in accordance with MIL-DTL-83513.

Plastic body or plastic molding: Shall conform to the requirements of GDI-30F or type SDG-F in accordance with ASTM D5948 or GCT-30F in accordance with ASTM D5927 or MIL-M-24519 or GST-40F in accordance with ASTM D4067 or MIL-M-24519 or GLCP-30F or GLCP-50 in accordance with ASTM D5138 or MIL-M-24519.

Jackpost: Corrosion resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with SAE-AMS-QQ-P-35.

Wire termination pins: Wire termination pins shall conform to A-A-59551, no. 24 AWG copper.

Mating connector: Shall conform to MIL-DTL-83513/1 and MIL-DTL-83513/3.

Plating of termination leads: Solder dipping of termination leads will be accomplished in SN60 PB40 or SN63 PB37 in accordance with J-STD-006.

Part or Identifying Number (PIN): PIN shall consist of the letter M, the basic number of the specification sheet, a letter from the insert, a numerical code for the termination length, and a letter code for the shell finish and hardware column.

M83513/15-	H	01	C	P
Specification sheet number	Insert arrangements (see figure 2) H = 100	Termination length 01 = .109 02 = .140 03 = .172	Shell finish (interface critical) C = cadmium N = electroless nickel (space application) P = passivated stainless steel	Hardware N = no jackpost or threaded insert P = jackpost attached T = threaded insert W = jackpost and threaded insert

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Referenced documents. In addition to MIL-DTL-83513, this document references the following:

MIL-DTL-83513/1
MIL-DTL-83513/3
MIL-M-24519
A-A-59551
ASTM A484
ASTM A582
ASTM D4067
ASTM D5138
ASTM D5927
ASTM B5948
J-STD-006
SAE-AMS-QQ-P-35

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AT, MI
Navy - AS, CG, MC, SH
Air Force – 99

Preparing activity:
DLA - CC

(Project 5935-4589-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsible can change, you should verify the currency of the information above using the ASSIST Online database at http://www.dodssp.daps.mil.

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